

PostgreSQL from Local to Cloud Docker, pgAdmin, and Supabase

CST 4714 – Database Administration

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Why This Lesson Matters

- Database admins work across environments
- Local $\neq$ cloud, but the database is the same
- Tools shape how we reason about systems

What Is Docker?

- Containerization platform
- Packages software + dependencies
- Not a virtual machine

Why Use Docker for Databases?

- Consistent versions
- Easy setup and teardown
- Reproducible environments

Installing Docker

- Download Docker Desktop
- Available for macOS, Windows, Linux
- Runs in the background

What Is Docker Compose?

- Defines multi-container systems
- Uses YAML configuration
- Infrastructure as code

Our docker-compose.yml

- PostgreSQL 15 image
- Environment variables
- Ports and volumes

Starting PostgreSQL with Docker

- `docker compose up -d`
- Runs in the background
- Creates database automatically

Connecting via the Terminal

- `psql` command-line client
- Direct access to PostgreSQL
- Works everywhere

Connecting via pgAdmin

- Graphical administration tool
- Schema browsing
- Query tool and statistics

Databases vs Schemas

- Database: top-level container
- Schema: namespace inside a database
- public is the default schema

Inspecting Tables and Stats

- information_schema
- pg_catalog
- Database size and activity

Practice Data: pgExercises

- Realistic relational schema
- Joins, aggregations, constraints
- Industry-relevant practice

From Local to Supabase

- Same PostgreSQL engine
- Managed infrastructure
- Browser-based tools

What Supabase Adds

- Automatic backups
- Authentication and RLS
- API layer

Local vs Cloud Responsibilities

- Docker: you manage everything
- Supabase: shared responsibility
- SQL remains unchanged

Key Takeaway

- PostgreSQL is portable
- Tools shape responsibility
- DBA thinking transcends platforms